Lab Record

Handwritten Lab Record Answers | JNTU(A) R23

Syllabus Topics:

(23A03304) MECHANICS OF SOLIDS & MATERIAL SCIENCE LAB - II Year B.Tech ME - I Semester - JNTUA R23. ; SECTION A: 1 Tensile | 2a Bending SS beam | 2b Cantilever | 3 Torsion | 4a Brinell | 4b Rockwell | 4c Vickers | 5 Springs | 6a Charpy | 6b Izod | 7 Punch shear | 8 Liquid penetration. ; SECTION B: common metallography procedure | 1 Pure metals | 2 Steels | 3 Cast irons | 4 Non-ferrous | 5 Heat treated steels | 6 Jominy. ; VIRTUAL LAB: 1 Principal stresses | 2 Izod | 3 Charpy | 4 Rockwell | 5 Vickers

Note:

All observation readings are SAMPLE values for a standard specimen so the calculation method is clear. In the lab, replace them with your own readings and recompute. Microstructure answers describe what to sketch.

SECTION A : MECHANICS OF SOLIDS LAB

Exp 1: Tensile Test on Mild Steel

Aim: To conduct a tension test on a mild steel specimen and find yield stress, ultimate stress, breaking stress, % elongation, % reduction in area and Young's modulus.

Apparatus: Universal Testing Machine (UTM), mild steel round specimen, vernier caliper, scale, punch / divider for marking gauge length, extensometer.

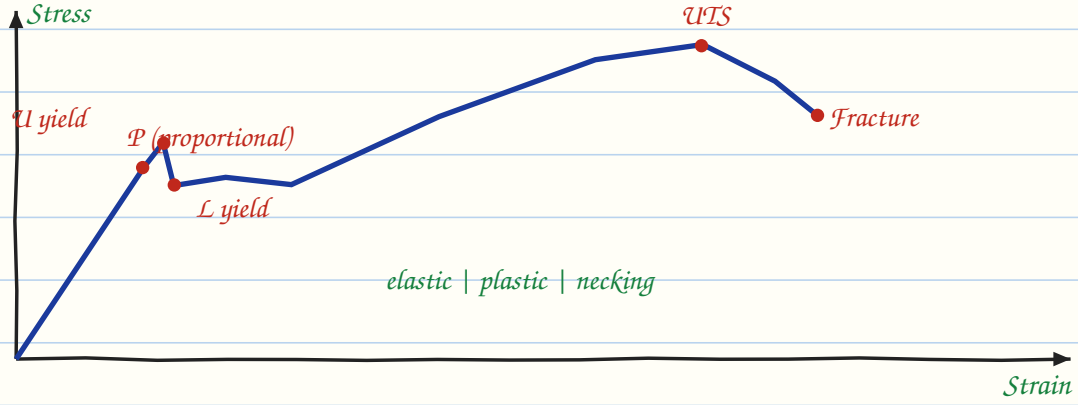


Fig: Stress-strain curve of mild steel

Theory

* Stress = P / A (N/mm2 = MPa); Strain = change in length / original length. Within the elastic limit stress is proportional to strain (Hooke's law): E = stress / strain.
* Mild steel shows a clear yield point (upper and lower), strain hardening up to ultimate load, then necking and fracture.

Yield stress = P_y / A0 ; UTS = P_max / A0 ; Breaking stress = P_b / A_f

% elongation = (L_f - L_0) / L_0 x 100 ; % reduction in area = (A_0 - A_f) / A_0 x 100

Procedure

1. Measure original diameter d0 and mark gauge length L0 = 5.65 sqrt(A0) (about 50 mm).
2. Grip the specimen in the jaws of UTM, set load and extension to zero, fit extensometer.
3. Apply load gradually; note load at each extension; note yield load (pointer halts or drops), maximum load and breaking load.
4. Remove the broken pieces, join them, measure final gauge length Lf and neck diameter df.

Observations (sample)

Quantity	Value
Original diameter d0	10 mm (A0 = 78.54 sq mm)
Gauge length L0	50 mm
Yield load Py	28.3 kN
Maximum load Pmax	40.0 kN
Breaking load Pb	30.0 kN
Final length Lf / neck dia df	63 mm / 6.5 mm (Af = 33.18 sq mm)

Calculations

* Yield stress = 28300 / 78.54 = 360 N/mm2. ; UTS = 40000 / 78.54 = 509 N/mm2. ; Breaking stress = 30000 / 33.18 = 904 N/mm2 (true).
* % elongation = (63 - 50)/50 x 100 = 26 %. ; % reduction in area = (78.54 - 33.18)/78.54 x 100 = 57.8 %.

Result: yield stress = 360 MPa, ultimate stress = 509 MPa, elongation = 26 %, reduction in area = 57.8 %, E about 2 x 10^5 MPa (from slope of the straight part).

Precautions: align the specimen axially, avoid slipping in the grips, apply load slowly, wear safety glasses.

Exp 2a: Bending Test - Simply Supported Beam

Aim: To determine Young's modulus E and the bending stress of a beam simply supported at ends and loaded at the centre.

Apparatus: Bending test setup with two knife-edge supports, hanger and weights, dial gauge, steel beam (25 mm x 10 mm), vernier, scale.

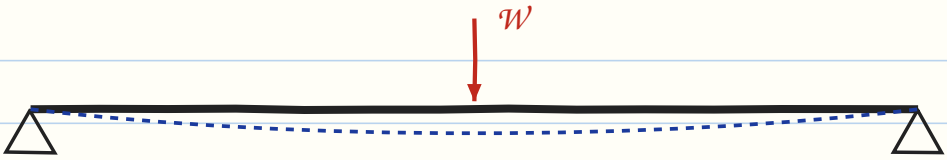


Fig: Simply supported beam with central load

Theory

Deflection at centre $d = W L^3 / (48 E I)$ -> $E = W L^3 / (48 I d)$

Bending equation: $M / I = \sigma / y = E / R$; $M_{max} = W L / 4$; $I = b d^3 / 12$

Procedure

1. Measure breadth b, depth d and span L of the beam; compute I.
2. Place the beam on the supports, fix dial gauge at the centre, set it to zero.
3. Add load in equal steps (20 N); note the deflection at each load; also take unloading readings.
4. Compute E for each reading and take the mean; compute bending stress.

Observations (sample): b = 25 mm, d = 10 mm, L = 800 mm, I = 2083.3 mm4

Load W (N)	Deflection d (mm)	E = 5120 W / d (GPa)
20	0.49	209.0
40	0.98	209.0
60	1.46	210.4
80	1.95	210.1

* Mean E = (209.0 + 209.0 + 210.4 + 210.1)/4 = 209.6 GPa (about 2.1 × 10^5 N/mm2).
* Bending stress for W = 20 N: M = 20 × 800/4 = 4000 N-mm; sigma = M y / I = 4000 × 5 / 2083.3 = 9.6 N/mm2.

Result: Young's modulus of the beam material E = 209.6 GPa; load-deflection graph is a straight line (elastic behaviour).

Exp 2b: Bending Test - Cantilever Beam

Aim: To find Young's modulus and the maximum bending stress of a cantilever beam by applying a load at the free end.

Apparatus: Cantilever test rig with clamp, hanger and weights, dial gauge, steel beam, vernier and scale.

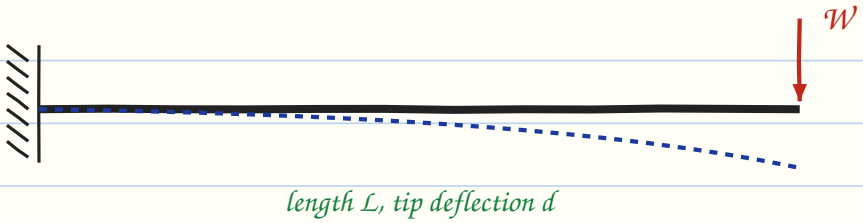


Fig: Cantilever with end load

Theory

$$\text{Tip deflection } d = W L^3 / (3 E I) \;\Rightarrow\; E = W L^3 / (3 I d)$$

$$M_{max} = W L \text{ (at fixed end)}; \quad \sigma_{max} = M_{max} y / I$$

Procedure

1. Clamp one end of the beam rigidly; measure free length L, b, d.
2. Fix the dial gauge below the free end and set to zero.
3. Apply loads 10, 20, 30, 40 N at the free end; note deflections.
4. Calculate E; plot W vs d (straight line).

Observations (sample): L = 500 mm, b = 25 mm, d = 10 mm, I = 2083.3 mm4

Load W (N)	Deflection d (mm)	E = 20000 W / d (GPa)
10	0.95	210.5
20	1.90	210.5
30	2.86	209.8
40	3.81	210.0

* Mean E = 210.2 GPa. ; For W = 40 N: M = 40 x 500 = 20000 N-mm, σ_{max} = 20000 x 5 / 2083.3 = 48.0 N/mm2.

Result: E of the cantilever material = 210.2 GPa; maximum bending stress at the fixed end for 40 N = 48 MPa.

Precautions: clamp firmly, keep dial gauge vertical, do not exceed elastic limit.

Exp 3: Torsion Test

Aim: To conduct a torsion test on a mild steel specimen and determine modulus of rigidity G and the maximum shear stress.

Apparatus: Torsion testing machine with torque and angle of twist scales, round steel specimen, vernier caliper, scale.

Theory

* A shaft under a twisting moment T develops shear stress that varies linearly from zero at the centre to maximum at the surface.

$$T / J = \tau / r = G\theta / L \qquad \text{(torsion equation)}$$

$$G = T L / (J \theta)\ ;\ J = \pi d^4 / 32\ ;\ \tau_{max} = 16 T / (\pi d^3)$$

theta must be in radians (1 deg = 0.01745 rad). Within the elastic limit T vs theta is a straight line; G = L / J × (slope).

Procedure

1. Measure diameter d and gauge length L of the specimen; fix it between the fixed and rotating heads.
2. Set torque and angle readings to zero; apply torque in equal steps by rotating the handwheel.
3. Note the angle of twist for every torque; continue up to fracture if required (ductile: large twist; brittle: 45 degree helical fracture).
4. Plot torque (y) vs angle of twist (x); calculate G.

Observations (sample): d = 10 mm, L = 200 mm, J = 981.7 mm4

Torque T (N-m)	Twist theta (deg)	theta (rad)	G = T L/(J theta) (GPa)
5	0.73	0.01274	79.9
10	1.46	0.02548	79.9

Torque T (N-m)	Twist theta (deg)	theta (rad)	G = T L/(J theta) (GPa)
15	2.19	0.03822	79.9
20	2.92	0.05096	79.9

* tau_max at T = 20 N-m = 16 x 20000 / (pi x 1000) = 101.9 N/mm2.

Result: modulus of rigidity G = 79.9 GPa (about 8 x 10^4 N/mm2) for mild steel; shear stress at 20 N-m = 101.9 MPa.

Exp 4a: Brinell Hardness Test

Aim: To determine the Brinell hardness number (BHN) of a given metal specimen.

Apparatus: Brinell hardness tester, hardened steel ball indenter D = 10 mm, specimen, Brinell microscope with graduated eyepiece.

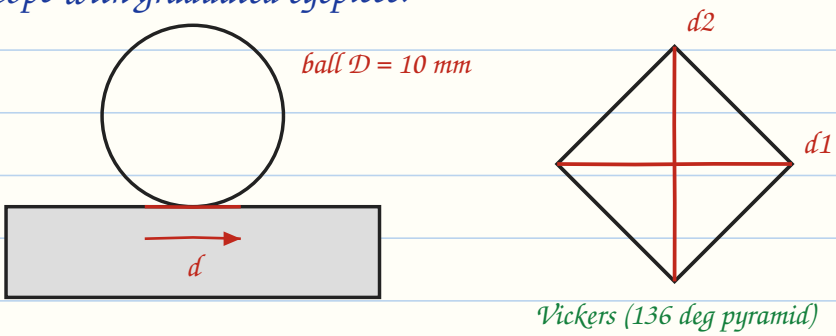


Fig: Brinell indentation (left) and Vickers indentation (right)

Theory

* Hardness = resistance to indentation / scratching / wear. In Brinell test a hardened steel (or tungsten carbide) ball is pressed with load P for 10 - 15 s; the diameter d of the permanent circular impression is measured.

$$BHN = 2P / (\pi D (D - \sqrt{D^2 - d^2}))$$ (kgf / sq mm)

* Standard loads: P = 30 D^2 (steels, cast iron) = 3000 kgf for D = 10 mm; 10 D^2 (copper alloys); 5 D^2 (aluminium alloys); 2.5 D^2 (soft metals). Specimen thickness at least 8 x depth of impression; indent distance from edge at least 2.5 d.

Procedure

1. Select ball and load as per the material; clean the specimen surface and place it on the anvil.

2. Raise the anvil to touch the ball; apply the load by the lever / hydraulic action for 15 s; release.
3. Measure two diameters of the impression at right angles using microscope; take the mean.
4. Calculate BHN; repeat three times at different places.

Observations (sample): mild steel, D = 10 mm, P = 3000 kgf

Trial	d1 (mm)	d2 (mm)	d mean (mm)	BHN
1	4.00	4.00	4.00	228.7
2	4.05	4.05	4.05	223.2
3	3.95	3.95	3.95	234.3

* Model: d = 4.0 -> sqrt(100 - 16) = 9.165; D - 9.165 = 0.835; pi × 10 × 0.835 = 26.23; BHN = 6000 / 26.23 = 228.7. Mean BHN = 228.7.

Result: Brinell hardness of the specimen = 229 BHN. (Approximate UTS of steel = 3.45 × BHN = 790 MPa.)

Precautions: smooth flat surface; apply the load smoothly; do not test on curved surface without correction.

Exp 4b: Rockwell Hardness Test

Aim: To determine the Rockwell hardness number of the given materials on B and C scales.

Apparatus: Rockwell hardness tester, diamond cone indenter (120 deg, for C scale), 1/16 inch steel ball indenter (B scale), test blocks.

Theory

* Rockwell test measures the depth of penetration. First a minor load (10 kgf) is applied to seat the indenter and set the zero; then the major load is applied and removed, leaving the minor load; the increase in depth is read directly on the dial as hardness number.

$$HR = N - h / 0.002 \quad (h = \text{permanent depth in mm};\ N = 100 \text{ for C}, 130 \text{ for B})$$

Scale	Indenter	Minor / Major load (kgf)	Used for
A	diamond cone	10 / 60	carbides, thin hard sheet
B	1/16 in ball	10 / 100	soft steel, brass, aluminium
C	diamond cone	10 / 150	hardened steel, HRC 20-70

* **Advantages over Brinell:** fast, direct reading, small indentation, suitable for finished parts, no optical measurement.

Procedure

1. Fit the indenter and select the major load for the scale; place a clean specimen on the anvil.
2. Raise the specimen until the minor load is applied; set the dial (C-scale pointer at 0, B-scale at 30).
3. Apply the major load by releasing the handle; after the pointer stops, remove the major load; read the dial.
4. Take 3 readings at different places (spacing at least 3 x indent dia.) and average.

Observations (sample)

Material	Scale	Reading 1	Reading 2	Reading 3	Mean HR
Mild steel	B	82	84	83	83 HRB
Brass	B	58	60	59	59 HRB

Material	Scale	Reading 1	Reading 2	Reading 3	Mean HR
Aluminium	B	44	46	45	45 HRB
Hardened steel	C	44	46	45	45 HRC

Result: Rockwell hardness: mild steel 83 HRB, brass 59 HRB, aluminium 45 HRB, hardened steel 45 HRC.

Exp 4c: Vickers Hardness Test

Aim: To find the Vickers hardness number (HV) of the given specimen.

Apparatus: Vickers hardness tester with square-based diamond pyramid indenter (136 degree between opposite faces), measuring microscope, specimen.

Theory

* The pyramid is pressed with load P (1 - 100 kgf) for 10 - 15 s; the two diagonals d1 and d2 of the square impression are measured; hardness = load / surface area of impression.

$$HV = 2P\sin(136/2) / d^2 = 1.854P / d^2$$

(d = (d1 + d2)/2 in mm, P in kgf)

* Advantages: one scale for all metals from soft to very hard; very small impression suitable for thin sheets, case-hardened layers, micro-hardness (HV 0.1 - 1 kgf).

Procedure

1. Polish the specimen surface; mount it on the stage; focus the microscope.
2. Select the load (e.g. 30 kgf); bring the indenter into position; apply load for 15 s and release.
3. Rotate the turret to the microscope; measure d1 and d2; take the mean.
4. Compute HV; take at least three readings.

Observations (sample): P = 30 kgf

Trial	d1 (mm)	d2 (mm)	d (mm)	HV = 1.854 P / d^2
1	0.40	0.42	0.41	330.9
2	0.41	0.41	0.41	330.9
3	0.42	0.42	0.42	315.3

* Model: HV = 1.854 x 30 / 0.41^2 = 55.62 / 0.1681 = 330.9. ; Mean HV = 325.7 (about 326 HV 30).

Result: Vickers hardness number of the specimen = 326 HV30. The load-time-diagonal values should be reported with the number, e.g. 326 HV30/15.

Exp 5: Test on Springs (Closed and Open Coil)

Aim: To determine the stiffness of a helical spring and the modulus of rigidity of the spring wire, and to grade the closed coil and open coil springs.

Apparatus: Spring testing machine / spring test rig with scale, hanger and weights, closed coil and open coil springs, vernier caliper, micrometer.

Theory

* Closed coil spring: coils are so close that the helix angle is small; the wire is mainly in torsion under an axial load W.

delta = 64 W R^3 n / (G d^4) ; stiffness k = W / delta = G d^4 / (64 R^3 n)

tau_max = 16 W R / (pi d^3) (Wahl factor K = (4C-1)/(4C-4) + 0.615/C for accuracy)

* Open coil spring: helix angle alpha is large; the axial load produces torsion and also bending in the wire, so the spring is less stiff: delta = (64 W R^3 n sec alpha / (G d^4)) (cos^2 alpha + (2G/E) sin^2 alpha); for small alpha it reduces to the closed-coil formula.
* R = mean coil radius, d = wire diameter, n = number of active coils, C = D / d (spring index).

Procedure

1. Measure wire diameter d, mean coil diameter D = 2R and count the active turns n.
2. Mount the spring vertically; note the initial scale reading with the hanger only.
3. Add loads in equal steps (10 N); note the extension each time; unload and repeat.

4. Plot load vs deflection; slope = stiffness k; calculate G = 64 k R^3 n / d^4.

Observations (sample, closed coil): d = 6 mm, D = 40 mm (R = 20 mm), n = 10

Load W (N)	Deflection (mm)	k = W / delta (N/mm)	G = 3950.6 k (GPa)
10	0.50	20.0	79.0
20	0.99	20.2	79.8
30	1.48	20.3	80.1
40	1.98	20.2	79.8

* Mean k = 20.2 N/mm; mean G = 79.7 GPa. Theoretical k = 80000 x 6^4 / (64 x 20^3 x 10) = 20.25 N/mm (agrees).

Result: stiffness of closed coil spring = 20.2 N/mm, G = 79.7 GPa. The open coil spring of same wire shows a smaller stiffness (larger deflection for the same load) because of added bending; so closed coil is graded stiffer.

Exp 6a: Charpy Impact Test

Aim: To determine the impact energy (toughness) of the given specimens using the Charpy test.

Apparatus: Charpy impact testing machine (pendulum 300 J), standard V-notch specimens 10 x 10 x 55 mm (notch 2 mm deep, 45 degree), vernier caliper.

Theory

* Impact test measures the energy absorbed in fracturing a notched bar under a sudden blow; it shows toughness and the ductile-brittle behaviour of the material.
* **Charpy:** specimen is supported as a simply supported beam (span 40 mm) with the notch on the side opposite the striking point; the pendulum released from height h hits at the centre.

$$\text{Energy absorbed } E = W R (\cos\beta - \cos\alpha) = W (h1 - h2)$$

(alpha = fall angle, beta = rise angle)

$$\text{Impact strength} = E / A \ (J / \text{sq mm}), \ A = \text{area under the notch} = 10 \times 8 = 80 \text{ sq mm}$$

* **Ductile fracture:** fibrous, large energy; brittle fracture: crystalline, small energy. Low temperature or high carbon content lowers the energy (ductile-brittle transition).

Procedure

1. Measure the specimen dimensions; raise the pendulum to the starting position and lock; set the pointer to maximum.
2. Place the specimen on the anvil with the notch facing away from the hammer, centred by the setting gauge.
3. Release the pendulum; after the swing apply the brake; read the energy absorbed on the dial.
4. Examine the fracture surface; repeat for other materials.

Observations (sample)

Material	Energy absorbed E (J)	Area (sq mm)	E / A (J/sq mm)	Fracture
Mild steel	92	80	1.15	fibrous (ductile)
Brass	38	80	0.48	mixed
Cast iron	4	80	0.05	brittle, granular

Result: Charpy impact energy: mild steel 92 J, brass 38 J, cast iron 4 J. Mild steel is the toughest; cast iron is brittle.

Exp 6b: Izod Impact Test

Aim: To determine the impact strength of the given metal specimens using the Izod test.

Apparatus: Izod impact tester (pendulum 168 J), standard specimen 10 x 10 x 75 mm with a 45 degree V-notch (2 mm deep), vice, vernier caliper.

Theory

* Izod: specimen is held vertically as a cantilever (clamped at one end) with the notch at the face of the vice on the side of the striker; the hammer strikes the free end 22 mm above the notch.
* Energy absorbed is found from the loss of potential energy of the pendulum as in Charpy test.

$$E = W R (\cos\beta - \cos\alpha)\ (J);\ \text{Izod strength} = E / \text{notch area}$$

Property	Charpy	Izod
Support	simply supported beam	cantilever (vertical)
Specimen	10 x 10 x 55 mm	10 x 10 x 75 mm
Notch position	opposite to hammer	facing hammer
Striking energy	300 J	168 J

Procedure

1. Raise the hammer and lock it; set the dial pointer to the maximum reading.
2. Clamp the specimen vertically with the notch exactly in line with the top of the vice (use the gauge).

3. Release the pendulum; the hammer breaks the specimen; apply the brake and read the energy.
4. Repeat for other materials; note the type of fracture.

Observations (sample)

Material	Energy absorbed (J)	Impact strength (J/sq mm)
Mild steel	85	1.06
Aluminium	22	0.28
Cast iron	3	0.04

Result: Izod impact energy: mild steel 85 J, aluminium 22 J, cast iron 3 J.

Precautions: the notch must be exactly at the vice face; keep away from the swing path of the pendulum; use the brake only after the swing.

Exp 7: Punch Shear Test

Aim: To determine the shear strength of a given sheet metal by punching a hole through it.

Apparatus: UTM with punch and die (shear) attachment, sheet specimens, vernier caliper / micrometer.

Theory

* When a punch of diameter d pushes through a sheet of thickness t, the plug is sheared out along the cylindrical surface of area pi d t. The maximum load recorded is the shear force.

$$\text{Shear strength } \tau = P / (\pi d t)$$

* **Empirical:** shear strength of steel about 0.6 - 0.75 x UTS; the same relation is used to find the force needed for punching, blanking and shearing operations in press working.

Procedure

1. Measure thickness t of the sheet and punch diameter d.
2. Place the sheet between the die plate and the punch holder on the UTM.
3. Apply the compressive load slowly until the punch cuts through; note the maximum load P.
4. Repeat for different materials; compute tau.

Observations (sample): d = 10 mm, t = 2 mm, shear area = pi x 10 x 2 = 62.83 sq mm

Material	Maximum load P (kN)	Shear strength (MPa)
Mild steel sheet	20.1	319.9
Brass sheet	13.2	210.1
Aluminium sheet	7.5	119.4

* **Model:** mild steel tau = 20100 / 62.83 = 319.9 N/mm2.

Result: shear strength of mild steel = 320 MPa, brass = 210 MPa, aluminium = 119 MPa.

Precautions: keep the punch and die concentric; use hardened punch; the plug should fall freely.

Exp 8: Liquid (Dye) Penetration Test

Aim: To detect surface-breaking defects (cracks, porosity, laps) in the given specimen by the liquid penetrant method of non-destructive testing.

Apparatus: Penetrant kit (cleaner, red visible dye penetrant, developer), lint-free cloth, specimen with a crack, white light / UV lamp (for fluorescent type).

Theory

* A low-viscosity, low surface-tension liquid enters surface-opening discontinuities by capillary action. After the excess is removed, a developer (white chalk-like powder) draws the trapped penetrant out by blotting, giving a visible indication larger than the defect.
* **Types:** Type I fluorescent, Type II visible dye. Methods: A water-washable, B post-emulsifiable, C solvent-removable (portable, most common).
* Detects only surface-open defects; works on non-porous materials (metals, ceramics, plastics) including non-magnetic metals where magnetic particle test cannot be used.

Procedure

1. Pre-clean the surface (solvent / wire brush) and dry; remove oil, rust and paint.
2. Apply penetrant by spray or brush; keep wet for the dwell time 5 - 20 minutes.
3. Remove excess penetrant with a dry cloth and then a cloth lightly dampened with cleaner (do not flood).
4. Apply a thin even layer of developer; wait 7 - 10 minutes.
5. **Inspect under good light (or UV):** defects appear as red lines / dots on the white background; record their location, length and type; post-clean the part.

Indication	Defect type
Continuous red line	crack, cold shut
Intermittent line	tight crack, lap, grinding crack
Round dots / clusters	porosity, pinholes
Broad bleed-out	wide open crack

Observation: a continuous red line 12 mm long was visible on the specimen at the weld toe, showing a surface crack (sample).

Result: the surface crack in the specimen has been detected using dye penetrant test.

Advantages: low cost, simple, portable, high sensitivity; limitation: cannot detect sub-surface defects and needs clean surface.

SECTION B : MATERIAL SCIENCE LAB

Common Procedure: Specimen Preparation (Metallography)

Aim: To prepare a metal specimen for microscopic examination. This procedure is common to experiments B1 to B5.

Apparatus: Hacksaw / cut-off wheel, mounting press (bakelite), belt grinder, emery papers 120, 220, 400, 600, 800, 1000, 1200 grit, polishing wheel with velvet cloth, alumina (0.3 / 0.05 micron) or diamond paste, etchant, alcohol, hot air dryer, metallurgical microscope.

1. **Sectioning:** cut a small piece (10 - 15 mm) with coolant, avoiding heating and distortion.
2. **Mounting:** for small samples, mount in bakelite or cold-setting resin.
3. **Coarse and fine grinding:** rub on successively finer emery paper, turning the specimen by 90 degrees after each paper until the old scratches vanish; use water.
4. **Polishing:** rough polish with alumina slurry on the wheel, final polish with 0.05 micron slurry/diamond paste until a mirror finish with no scratches is obtained; wash with water and alcohol; dry.
5. **Etching:** immerse or swab in a suitable etchant for the correct time; it attacks grain boundaries and phases at different rates; wash and dry immediately.
6. **Observation:** place on the microscope stage; use 100x, 200x, 500x; sketch the structure and mark the phases.

Material	Etchant
Steels, cast iron	2 - 4 % Nital (nitric acid in alcohol); picral for fine pearlite
Aluminium alloys	Keller's reagent / 0.5 % HF
Copper, brass	ammonical ferric chloride
Pure iron / ferrite	2 % Nital

Precautions: do not press hard while polishing (to avoid relief and drag), etch for only a few seconds, handle acids with gloves, never touch the polished face.

Exp B1: Microstructure of Pure Metals

Aim: To prepare and study the microstructure of pure metals such as pure iron, copper and aluminium.

Apparatus: As in the common procedure; specimens of pure Fe (ARMCO iron), Cu and Al; etchants: Nital (Fe), ammonical FeCl3 (Cu), Keller's reagent (Al).

Theory

* A pure metal solidifies as a single phase made up of many crystals (grains), each of a fixed lattice orientation. The boundary between grains is the grain boundary - a region of mismatch which etches faster and shows as dark lines.
* Grain size depends on cooling rate and heat treatment; fine grain gives higher strength (Hall-Petch relation: sigma_y = sigma_0 + k / sqrt(d)). ASTM grain size number G: N = 2^(G-1) grains per sq inch at 100x.

Metal	Crystal structure	Structure seen
Pure iron	BCC (alpha-ferrite)	polygonal equiaxed ferrite grains, boundaries dark
Copper	FCC	equiaxed grains with straight annealing twins
Aluminium	FCC	equiaxed grains, light, some dark pits

Procedure

* Prepare the specimens as in the common procedure, etch each with its etchant (Fe 10 - 15 s, Cu 20 s, Al 15 s), observe at 100x - 200x and sketch.

Observation (to sketch): irregular polygonal grains separated by thin dark boundaries; in copper straight parallel lines inside grains are annealing twins; no second phase is seen.

Result: the microstructure of pure metals consists of single-phase equiaxed grains with grain boundaries; the average grain size was found by the intercept method: d = L / (N M) (L = test line length, N = intersections, M = magnification).

Exp B2: Microstructure of Mild, Medium and High Carbon Steels

Aim: To prepare and study the microstructures of mild steel, medium carbon steel and high carbon steel, and to estimate their phase percentages.

Apparatus: Specimens of 0.2 % C, 0.45 % C, 0.8 % and 1.2 % C steels, polishing set, 2 % Nital, metallurgical microscope.

Theory (Fe - Fe3C diagram, slow cooled)

* **Ferrite (alpha):** soft, ductile, appears white/light; Pearlite: lamellar eutectoid mixture of ferrite and cementite (0.8 % C at 727 C), appears dark/grey with fine lines; Cementite (Fe3C): hard brittle, white network at grain boundaries in hypereutectoid steel.

$$\% \text{ pearlite} = (C - 0.02) / (0.8 - 0.02) \times 100 \; ; \quad \% \text{ ferrite} = 100 - \% \text{ pearlite}$$

(hypoeutectoid, lever rule)

Steel	% C	Structure	% pearlite	Properties
Mild steel	0.2	ferrite grains + islands of pearlite	23	soft, ductile, weldable
Medium carbon	0.45	about equal ferrite and pearlite	54	strong, tough
Eutectoid steel	0.8	100 % pearlite (lamellar)	100	strong, hard
High carbon	1.2	pearlite + cementite network at boundaries	~ 95 + 5 % Fe3C	very hard, brittle

Procedure and observation

* Prepare the four specimens and etch with Nital for 5 - 10 s; observe at 100x and 400x. Draw: (a) white ferrite with dark pearlite patches, (b) more pearlite with ferrite network, (c) fully dark lamellar pearlite, (d) pearlite surrounded by white cementite network.
* **Model:** 0.45 % C: % pearlite = (0.45 - 0.02) / 0.78 = 55 % (approx. 54 %); ferrite 45 %.

Result: As carbon increases from 0.2 to 1.2 %, pearlite content increases, so hardness and strength increase and ductility decreases.

Exp B3: Microstructure of Cast Irons

Aim: To study the microstructures of grey, white, nodular (SG) and malleable cast irons.

Apparatus: Specimens of the four cast irons, polishing set, 2 % Nital, metallurgical microscope.

Theory

* **Cast iron:** Fe - C alloy with 2.1 - 4.5 % C and 1 - 3 % Si; carbon occurs as graphite or as cementite depending on composition and cooling rate (silicon and slow cooling promote graphite).

Type	Structure	Properties / uses
Grey CI	graphite FLAKES in pearlite / ferrite matrix (unetched: dark flakes)	good damping, machinability, compression strength; engine blocks, bed of machine tools
White CI	cementite + pearlite (ledeburite), no graphite; white fracture	very hard, brittle, wear resistant; liners, rolls
Nodular (SG) CI	graphite NODULES (bull's-eye: ferrite ring around the nodule)	high strength and ductility (Mg / Ce treated); crankshafts, gears
Malleable CI	temper carbon ROSETTES in ferrite / pearlite	tough, ductile; pipe fittings, automobile parts

Procedure

* Prepare as in the common procedure; avoid pulling out graphite while polishing; observe first unetched (graphite shape and distribution) and then after etching (matrix).

Result: The graphite shapes (flakes, nodules, rosettes) and matrix structures were identified; the properties of cast iron are controlled by the shape of the graphite and the matrix.

Exp B4: Microstructure of Non-Ferrous Alloys

Aim: To study the microstructures of non-ferrous alloys: brass, bronze and aluminium alloys.

Apparatus: Specimens of alpha-brass (70/30), alpha+beta brass (60/40 Muntz metal), tin bronze, Al-Si alloy; polishing set; etchants; microscope.

Alloy	Etchant	Microstructure
70/30 cartridge brass (alpha)	ammonical FeCl3	single phase alpha grains with annealing twins
60/40 Muntz metal (alpha + beta)	ammonical FeCl3	light alpha with dark beta phase
Tin bronze (cast)	FeCl3	cored dendrites of alpha with alpha + delta eutectoid
Al - Si (12 % Si, eutectic)	0.5 % HF	needles / flakes of silicon in aluminium matrix; modified by sodium to fine fibrous
Al - 4.5 % Cu	Keller's	alpha solid solution with CuAl2 at the boundaries

* **Brass (Cu - Zn):** alpha is soft, ductile (cold-working); beta is harder, good for hot work. Bronze (Cu - Sn): wear and corrosion resistant (bearings, bushes). Aluminium alloys: light weight, castable (pistons: Al - Si).

Result: Alloy phases and their distribution were studied; the observed structures agree with the respective phase diagrams.

Exp B5: Microstructure of Heat Treated Steels

Aim: To study the microstructures of steel after annealing, normalising, hardening and tempering.

Apparatus: Muffle furnace, tongs, quenching media (water, oil), medium carbon steel specimens (0.45 % C), polishing set, 2 % Nital, microscope.

Procedure

1. Heat the specimens to 830 - 850 C (30 - 50 C above A3) and soak for about 30 min (1 hr per 25 mm).
2. **Cool:** furnace cool (annealed), air cool (normalised), oil quench, water quench (hardened); re-heat some quenched samples to 200 - 600 C (tempered).
3. Prepare, etch and observe each sample; measure hardness if possible.

Treatment	Cooling	Microstructure	Hardness (HRC approx.)
Annealing	furnace	coarse pearlite + ferrite	15 - 20
Normalising	air	fine pearlite + ferrite (fine grains)	20 - 28
Oil quench	oil	martensite + some bainite / pearlite	45 - 55
Water quench	water	needle-like (acicular) martensite	55 - 60
Tempered (400 C)	after quench	tempered martensite (troostite)	40 - 45

* **Martensite:** supersaturated BCT solid solution of carbon in iron formed by diffusionless transformation; hard and brittle. Tempering relieves stresses and increases toughness.

Result: The cooling rate decides the microstructure: slow cooling gives soft pearlite, fast cooling gives hard martensite, and tempering gives tempered martensite (hardness decreases, toughness increases).

Exp B6: Hardenability of Steels by Jominy End Quench Test

Aim: To determine the hardenability of a given steel by the Jominy end quench test.

Apparatus: Jominy test bar 25 mm dia x 100 mm long, furnace, Jominy quench fixture with water jet (free height 65 mm, orifice 12.5 mm), surface grinder, Rockwell hardness tester.

Theory

* **Hardenability:** ability of steel to be hardened to a depth on quenching (not the maximum hardness; maximum hardness depends on carbon content). It depends on alloying elements (Mn, Cr, Mo, Ni) and austenite grain size.
* **Hardenability curve:** hardness vs distance from the quenched end; a flatter curve means higher hardenability (deep hardening).

Procedure (ASTM A255)

1. Heat the bar to the austenitising temperature (845 C for 0.45 % C steel) and hold 30 min.
2. Transfer it to the fixture within 5 seconds; quench with a water jet at the bottom end only for at least 10 min.
3. Grind two parallel flats 0.4 mm deep along the length; measure HRC at 1.5 mm intervals near the quenched end and 3 - 5 mm farther.
4. Plot HRC vs distance from the quenched end.

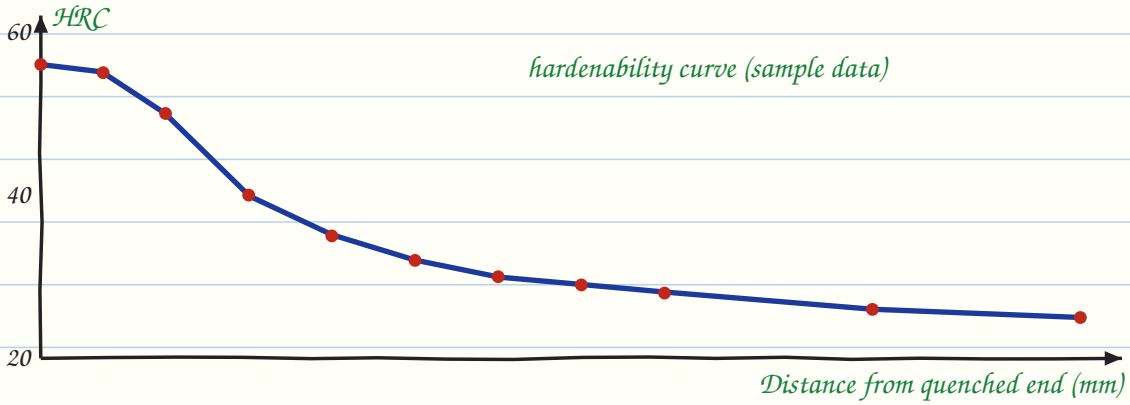


Fig: Jominy hardenability curve (sample)

Distance (mm)	0	1.5	3	5	7	9	11	15	25
HRC (sample)	56	55	50	40	35	32	30	28	25

Result: hardness falls from 56 HRC at the quenched end to 25 HRC at 25 mm; the 50 HRC (about 50 % martensite) point is at 3 mm, showing medium hardenability. An alloy steel would show a flatter curve.

Virtual Lab

Virtual labs are simulations on the web (links given in the syllabus). Open the link, read the Theory and Procedure tabs, then use the Simulator tab, record the readings and answer the quiz. Sample results are given below.

Virtual 1: Principal Stresses at a Point (Plane Stress)

Aim: To find the principal stresses sigma_a and sigma_b (and maximum shear stress) at a point of a structural element in a state of plane stress.

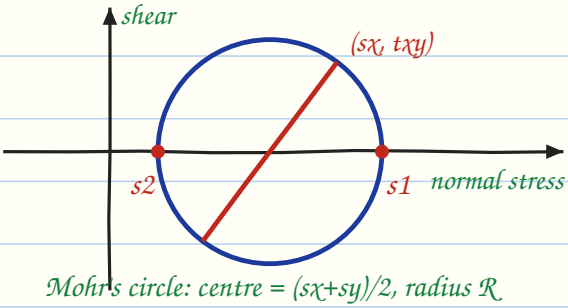


Fig: Mohr's circle

$$\sigma_{1,2} = \frac{(s_x + s_y)}{2} \pm \sqrt{\left(\frac{(s_x - s_y)}{2}\right)^2 + t_{xy}^2}$$

$$\tan 2\theta_p = \frac{2 t_{xy}}{(s_x - s_y)}; \quad \tau_{max} = \frac{(\sigma_1 - \sigma_2)}{2}$$

* Procedure: enter sx, sy, txy in the simulator; note principal stresses, the angle theta_p of principal planes and the Mohr circle; vary the inputs and verify by calculation.
* Example: sx = 80, sy = 20, txy = 40 MPa: centre = 50; R = sqrt(30^2 + 40^2) = 50; sigma_1 = 100 MPa, sigma_2 = 0 MPa; theta_p = 0.5 atan(80/60) = 26.57 deg; tau_max = 50 MPa.

Result: principal stresses sigma_a = 100 MPa and sigma_b = 0 MPa acting on planes at 26.57 degrees to the x-axis; maximum shear stress 50 MPa.

Virtual 2: Izod Impact Test (Mild Steel and Cast Iron)

Aim: To find the impact resistance of mild steel and cast iron using the Izod impact virtual lab.

* *Procedure:* choose the specimen (mild steel / cast iron) in the simulator; set the pendulum initial angle; release; note the final swing angle and the energy absorbed shown; repeat for the second material and compare.

$$E = W R (\cos \text{beta} - \cos \text{alpha})$$

Material	Initial angle alpha	Final angle beta	Energy (J)	Remark
Mild steel	150	90	high (about 80 J)	ductile, fibrous
Cast iron	150	140	low (about 8 J)	brittle

Result: mild steel absorbs far more energy than cast iron, so its impact resistance (toughness) is much higher.

Virtual 3: Charpy Impact Test (Mild Steel)

Aim: To find the impact resistance of mild steel by the Charpy virtual lab.

* *Procedure:* select mild steel specimen and temperature; place the specimen as a simply supported beam; release the pendulum; record the energy absorbed; repeat at different temperatures if the simulator allows to see the ductile-brittle transition.

Temperature (C)	Energy absorbed (J)	Fracture
27	92	ductile
0	60	mixed
-40	15	brittle

Result: impact resistance of mild steel at room temperature = 92 J; the energy falls sharply at low temperature (ductile-brittle transition).

Virtual 4: Rockwell Hardness of Different Materials

Aim: To find the Rockwell hardness number of mild steel, cast iron, brass, aluminium and spring steel using the virtual lab.

* **Procedure:** select the material; choose the scale / indenter as the simulator suggests (B scale: 1/16 inch ball, 100 kgf; C scale: diamond, 150 kgf); apply minor load, then major load; read the number; repeat three times.

Material	Scale	Hardness (sample)
Mild steel	B	83 HRB
Cast iron	B	95 HRB
Brass	B	59 HRB
Aluminium	B	45 HRB
Spring steel (hardened)	C	46 HRC

Result: hardness order: spring steel > cast iron > mild steel > brass > aluminium.

Virtual 5: Vickers Indentation Hardness

Aim: To determine the indentation hardness of mild steel, brass, aluminium, etc. using a virtual Vickers hardness tester.

* *Procedure:* select material and load (e.g. 30 kgf); press the indenter; read diagonals d1 and d2 from the microscope view; compute $HV = 1.854\ P / d^2$.

Material	P (kgf)	d (mm)	$HV = 1.854\ P / d^2$
Mild steel	30	0.50	222
Brass	30	0.68	120
Aluminium	30	0.90	69

* *Model:* mild steel $HV = 1.854 \times 30 / 0.25 = 222.5$.

Result: Vickers hardness mild steel 222 HV30, brass 120 HV30, aluminium 69 HV30.